Chapter 33

Campbell Biology, AP Edition, 12e

Photo of blue dragon (Glaucus atlanticus) floating in water shows a bright blue body that tapers to a point, with spiderlike appendages.

Chapter 33

An Introduction to Invertebrates

Figure 33.1

The blue dragon (*Glaucus atlanticus*) has thin, finger-like structures that help it to float (upside down) on the sea's surface. It eats Portuguese men-of-war and absorbs their stinging cells, which it then uses to deliver a deadly sting of its own. The blue dragon is just one of more than a million species of invertebrates (animals without a backbone), which account for over 95% of animal species.

Tree diagram showing animal phylogeny starting with ancestral protist.

Tree diagram for animal phylogeny represented as a list.

Ancestral Protist = Common ancestor of all animals (Branch Point 1). Porifera (sponge drawn).

(Eumetazoa) Branch point 2. Cnidaria (Hydra drawn).

(Bilateria) Branch point 3

Branch point 4. Lophotrochozoa (snail drawn). Ecdysozoa (crayfish drawn). Deuterostomia (fish drawn).

Key Concepts

- **33.1Sponges are basal animals that lack tissues****
- **33.2Cnidarians are an ancient phylum of eumetazoans****

- **33.3Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms****
- **33.4Ecdysozoans are the most species-rich animal group****
- **33.5Echinoderms and chordates are deuterostomes****

Advanced placement edition

Through natural selection, invertebrates have adapted to almost every habitat on Earth **(BIG IDEA 1)**. Invertebrates account for the majority of known animal species, and they interact with virtually all organisms on Earth **(BIG IDEA 4)**.

Study Tip

Make a table: Invertebrates differ in how they obtain food, remove wastes, reproduce, and move throughout their habitat. As you read, summarize how members of the 11 phyla of invertebrates described in detail in this chapter solve these shared challenges of life.

A table lists phylum, feeding method, waste removal, reproduction, and movement.

The table is as follows:

Phylum	Feeding method	Waste removal	Reproduction	Movement
Porifera	Filter feeders	Diffusion	Hermaphrodites	Sessile
blank	blank	blank	blank	blank

****Invertebrates** are animals that lack a backbone. They occupy almost every habitat on Earth, from the scalding water released by deep-sea hydrothermal vents to the frozen ground of Antarctica. Evolution in these varied environments has produced an immense diversity of forms, including species that are microscopic and species whose members can grow to 18 m long (1.5 times the length of a school bus). In this chapter, we'll take a tour of the invertebrates, using the phylogenetic tree in Figure 33.1 as a guide. **Figure 33.2** surveys 23 phyla as representatives of invertebrate diversity.

Figure 33.2

Exploring Invertebrate Diversity

Exploring invertebrate and vertebrate diversity.

Kingdom Animalia encompasses 1.3 million known species, and estimates of the total number of species range as high as 10–20 million. Of the 23 phyla surveyed here, 13 are discussed more fully in this chapter, Chapter 32, or Chapter 34; cross-references are given at the end of their descriptions.

Porifera (5,500 species): Animals in this phylum are informally called sponges. Sponges are sessile animals that lack tissues. They live as filter feeders, trapping particles that pass through the internal channels of their body (see Concept 33.1). A photograph of a sponge is shown.

Placozoa (1 species): The single known species in this phylum, Trichoplax adhaerens, doesn't look like an animal. It consists of a simple bilayer of a few thousand cells and reproduces by dividing into two individuals or by budding off many multicellular individuals. Placozoans are thought to be basal animals, but it is not yet known how they are related to Porifera, Cnidaria, and other early-diverging phyla. A light microscope image of a placozoan is shown and a scale bar reads 0.5 m m.

Cnidaria (10,000 species): Cnidarians include corals, jellies, and hydras. These animals have a diploblastic, radially symmetrical body plan that includes a gastrovascular cavity with a single opening that serves as both mouth and anus (see Concept 33.2). A photograph of a jelly is shown.

Acoela (400 species): Flatworms in this phylum have a simple nervous system and a saclike gut, and thus were once placed in phylum Platyhelminthes. Recent molecular analyses, however, indicate that Acoela is a separate lineage that diverged before the three main bilaterian clades (see Concept 32.4). Acoela organisms are shown with a scale bar reading 1.5 m m.

Ctenophora (100 species): Ctenophores (comb jellies) are diploblastic and radially symmetrical like cnidarians, suggesting that both phyla diverged early from other animals. Comb jellies make up much of the ocean's plankton. They have eight "combs" of cilia that propel the animals through the water. When a small animal contacts the tentacles of some comb jellies, specialized cells burst open, covering the prey with sticky threads. A photograph of a ctenopbore, or comb jelly is shown.

Lophotrochozoa

- Platyhelminthes (20,000 species): Flatworms in this phylum (including tapeworms, planarians, and flukes) have bilateral symmetry and a central nervous system that processes information from sensory structures. They have no body cavity or specialized organs for circulation (see Concept 33.3). A marine flatworm is shown.
- Syndermata (2,900 species): This recently established phylum includes two groups formerly classified as separate phyla: the rotifers, microscopic animals with complex organ systems, and the acanthocephalans, highly modified parasites of vertebrates (see Concept 33.3). A light microscope image of a rotifer is shown and a scale bar reads 100 micrometers.

- Ectoprocta (4,500 species): Ectoprocts (also known as bryozoans) live as sessile colonies and are covered by a tough exoskeleton (see Concept 33.3). An ectoproct is shown.
- Brachiopoda (335 species): Brachiopods, or lamp shells, may be easily mistaken for clams or other mollusks. However, most brachiopods have a unique stalk that anchors them to their substrate, as well as a crown of cilia called a lophophore (see Concept 33.3). A brachiopod is shown.
- Gastrotricha (800 species): Gastrotrichs are tiny worms whose ventral surface is covered with cilia, leading them to be called hairy bellies. Most species live on the bottoms of lakes or oceans, where they feed on small organ isms and partially decayed organic matter. This individual has consumed algae, visible as the greenish material inside its gut. A differential interference contrast light microscope image of a gastrotrich is shown and a scale bar reads 200 micrometers.
- Cycliophora (1 species): The only known cycliophoran species, Symbion pandora, was discovered in 1995 on the mouthparts of a lobster. This tiny creature has a unique body plan and a bizarre life cycle: Males impregnate females that are still developing in their mothers' bodies. The fertilized females escape, settle elsewhere on the lobster, and release their offspring. A colorized scanning electron microscope image of a cycliophoran is shown and a scale bar reads 100 micrometers.
- Nemertea (900 species): Also called ribbon worms, nemerteans swim through water or burrow in sand, extending a unique proboscis to capture prey. Nemerteans have a reduced coelom, so their bodies are relatively solid. They also have an alimentary canal and a closed circulatory system in which the blood is contained in vessels and hence is distinct from fluid in the body cavity. A photograph of a ribbon worm is shown.
- Annelida (16,500 species): Annelids, or segmented worms, are distinguished by their body segmentation. Earthworms are the most familiar annelids, but most annelids live in marine or freshwater habitats (see Concept 33.3). A photograph of a marine annelid is shown.
- Mollusca (100,000 species): Mollusks (including snails, clams, squids, and octopuses) have a soft body that in many species is protected by a hard shell (see Concept 33.3). A photograph of an octopus is shown.

Ecdysozoa

- Loricifera (10 species): Loriciferans (from the Latin lorica, corset, and ferre, to bear) are tiny animals that inhabit sediments on the seafloor. A loriciferan can telescope its head, neck, and thorax in and out of the lorica, a pocket formed by six plates surrounding the abdomen. Though the natural history of loriciferans is mostly a mystery, at least some species likely eat bacteria. A light microscope image of a loriciferan is shown and a scale bar reads 50 micrometers.
- Priapula (16 species): Priapulans are worms with a large, rounded proboscis at the anterior end. (They are named after Priapos, the Greek god of fertility, who was symbolized by a giant penis.)

Ranging from 0.5 mm to 20 cm in length, most species burrow through seafloor sediments. Fossil evidence suggests that priapulans were among the major predators during the Cambrian period. A photograph of a priapulan is shown.

- Onychophora (110 species): Onychophorans, also called velvet worms, originated during the Cambrian explosion (see Concept 25.3). Originally, they thrived in the ocean, but at some point they succeeded in colonizing land. Today they live only in humid forests. Onychophorans have fleshy antennae and several dozen pairs of saclike legs. A photograph of an onychophoran is shown.
- Tardigrada (800 species): Tardigrades (from the Latin tardus, slow, and gradus, step) are sometimes called water bears for their overall shape and lumbering, bearlike gait. Most tardigrades are less than 0.5 mm in length. Some live in oceans or fresh water, while others live on plants or animals. Harsh conditions may cause tardigrades to enter a state of dormancy; while dormant, they can survive for days at temperatures as low as –200 degrees C! A recent phylogenomic study found that over 15% of tardigrade genes entered their genome by horizontal gene transfer, the largest fraction known for any animal. A scanning electron microscope image of a tardigrade is shown and a scale bar reads 100 micrometers.
- Nematoda (25,000 species): Also called roundworms, nematodes are enormously abundant and diverse in the soil and in aquatic habitats; many species parasitize plants and animals. Their most distinctive feature is a tough cuticle that coats their body (see Concept 33.4). A photograph of a roundworm is shown.
- Arthropoda (1,000,000 species): The vast majority of known animal species, including insects, crustaceans, and arachnids, are arthropods. All arthropods have a segmented exoskeleton and jointed appendages (see Concept 33.4). A photograph of an arachnid, a spider is shown.

Deuterostomia

- Hemichordata (85 species): Like echinoderms and chordates, hemichordates are members of the deuterostome clade (see Figure 32.11). Hemichordates share some traits with chordates, such as gill slits and a dorsal nerve cord. The largest group of hemichordates is the enteropneusts, or acorn worms. Acorn worms are marine and generally live buried in mud or under rocks; they may grow to more than 2 m in length. A photograph of an acorn worm is shown.
- Chordata (60,000 species): More than 90% of all known chordate species have backbones (and thus are vertebrates). However, the phylum Chordata also includes two groups of invertebrates: lancelets and tunicates. See Chapter 34 for a full discussion of this phylum. A photograph of a tunicate is shown.

- Echinodermata (7,000 species): Echinoderms, such as sand dollars, sea stars, and sea urchins, are marine animals in the deuterostome clade that are bilaterally symmetrical as larvae but not as adults. They move and feed by using a network of internal canals to pump water to different parts of their body (see Concept 33.5). A photograph of a sea urchin is shown.

Get Ready for This Chapter

In this chapter, you will learn about the diversity of invertebrates, focusing on the evolutionary history and natural history of these organisms. Before beginning this chapter, you should be able to do the following:

- Describe key characteristics of bilaterian animals (see Concept 32.4).
- Identify a lineage's basal taxon in a phylogenetic tree (see Concept 26.1, Figure 26.5).
- Distinguish among terms related to the body cavity: coelom, hemocoel, acoelomate (see Concept 32.3, Figure 32.9).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following statements about bilaterian animals is *true*?

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All bilaterians are triploblastic (have three germ layers).
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That's correct.

Correct! If you had trouble with this question, review the following material: **Bilaterian animals** form an enormous clade whose members all have tissues (unlike sponges) as well as a two-sided or bilaterally symmetric form (unlike sponges and cnidarians). All bilaterally symmetric animals also are triploblastic. Bilaterians include both invertebrate and vertebrate animals.

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All bilaterians have radial symmetry.
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That's incorrect. Try again.

Incorrect. Review the following material: **Bilaterian animals** form an enormous clade whose members all have a two-sided or bilaterally symmetric form (unlike sponges and cnidarians). All bilaterally symmetric animals also are triploblastic. Bilaterians include both invertebrate and vertebrate animals.

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Most bilaterians have tissues, but some do not.
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That's incorrect. Try again.

Incorrect. Review the following material: **Bilaterian animals** form an enormous clade whose members all have a two-sided or bilaterally symmetric form (unlike sponges and cnidarians). All bilaterally symmetric animals also are triploblastic. Bilaterians include both invertebrate and vertebrate animals.

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All bilaterians are invertebrates.
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That's incorrect. Try again.

Incorrect. Review the following material: **Bilaterian animals** form an enormous clade whose members all have a two-sided or bilaterally symmetric form (unlike sponges and cnidarians). All bilaterally symmetric animals also are triploblastic. Bilaterians include both invertebrate and vertebrate animals.

Series of nested horizontal forked branches leading to group names; from the top the branch endpoints are Porifera, Ctenophora, Cnidaria, Acoela, Deuterostomia, Lophotrochozoa, and Ecdysozoa.

Diagram showing a series of horizontal forked branches. Reading left to right, the first branch point forks into a horizontal line leading to Porifera and another horizontal line, which branches again. This second branch point forks into a horizontal line leading to Ctenophora and another horizontal line, which branches again. This third branch point forks into a horizontal line leading to Cnidaria and another horizontal line, which branches again. This fourth branch point forks into a horizontal line leading to Acoela and another horizontal line, which branches again. This fifth branch point forks into a horizontal line leading to Deuterostomia and another horizontal line, which branches again into two lines leading to Lophotrochozoa and Ecdysozoa. At the far right of the diagram a vertical bar extending from Ecdysozoa through Acoela is labeled "Bilateria (most animals)." A vertical bar extending from Ecdysozoa through Ctenophora is labeled "Eumetazoa," and another vertical bar extending from Ecdysozoa through Porifera is labeled "Metazoa."

2. Based on the phylogenetic tree shown here, identify the basal taxon of metazoans (animals).

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Ecdysozoa
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That's incorrect. Try again.

Incorrect. Review the following material: The term **basal taxon** refers to a lineage that diverges from all other members of its group early in the history of the group. Hence, a basal taxon lies on a branch that originates near the common ancestor of the group.

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Eumetazoa
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That's incorrect. Try again.

Incorrect. Review the following material: The term **basal taxon** refers to a lineage that diverges from all other members of its group early in the history of the group. Hence, a basal taxon lies on a branch that originates near the common ancestor of the group.

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Acoela
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That's incorrect. Try again.

Incorrect. Review the following material: The term **basal taxon** refers to a lineage that diverges from all other members of its group early in the history of the group. Hence, a basal taxon lies on a branch that originates near the common ancestor of the group.

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Porifera
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That's correct.

Correct! If you had trouble with this question, review the following material: The term **basal taxon** refers to a lineage that diverges from all other members of its group early in the history of the group. Hence, a basal taxon lies on a branch that originates near the common ancestor of the group. In this phylogenetic tree, Porifera is shown as the first group of metazoans (animals) to have diverged from all other metazoans, and so Porifera is the basal taxon of metazoans.

3. A hemocoel refers to a

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functional body cavity lined by tissues derived from mesoderm and endoderm.
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That's correct.

Correct! If you had trouble with this question, review the following material: Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes. Many bilaterian animals have a coelom, a body cavity that forms from tissue derived from mesoderm. Other bilaterian animals have a hemocoel, a body cavity lined with tissue derived from mesoderm and with tissue derived from endoderm.

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        body cavity that does not function properly.
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That's incorrect. Try again.

Incorrect. Review the following material: Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes. Many bilaterian animals have a coelom, a body cavity that forms from tissue derived from mesoderm. Other bilaterian animals have a hemocoel, a body cavity lined with tissue derived from mesoderm and with tissue derived from endoderm.

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        functional body cavity lined with tissues derived only
from mesoderm.
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That's incorrect. Try again.

Incorrect. Review the following material: Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes. Many bilaterian animals have a coelom, a body cavity that forms from tissue derived from mesoderm. Other bilaterian animals have a hemocoel, a body cavity lined with tissue derived from mesoderm and with tissue derived from endoderm.

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        functional body cavity lined with tissues derived only
from endoderm.
```

That's incorrect. Try again.

Incorrect. Review the following material: Body cavities have diverse functions, such as to provide structural support and to facilitate the internal transport of nutrients, gases, and wastes. Many bilaterian animals have a coelom, a body cavity that forms from tissue derived from mesoderm. Other bilaterian animals have a hemocoel, a body cavity lined with tissue derived from mesoderm and with tissue derived from endoderm.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 33.1Describe the structure of the sponge body.
- 33.2Characterize cnidarians and their life cycles.
- 33.3Give examples of lophotrochozoans and describe their characteristics.
- 33.4Give examples of ecdysozoans and describe their characteristics.
- 33.5Identify major phylogenetic groups of deuterostomes and describe their characteristics.